SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	**CSA10 / CSA1005**	**Course Title:**	**Software Engineering**
CO Assessed:	**CO1**	**Assessment:**	**Assessment Tool 1 – Case Study Analysis**
Weightage:	**20%**	**Total Marks:**	**25**
CO1 Statement:	***Apply advanced methodologies and agile project management techniques to manage software projects effectively***		
Student Name:	**Mathivadhani.K**	**Reg. No.:**	**192311424**
Date:	**10.10.2026**	**Duration:**	**60 minutes**

Code of Conduct

I __Mathivadhani.K(192311424)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Mathivadhani.K__

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted one problem statement. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. Identify Key Stakeholders and Challenges
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. Analyze the Current Approach
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. Propose a Suitable Solution Using Software Engineering Principles
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. Justify the Chosen Methodology/Framework
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
   - Justify why the selected methodology/framework is suitable for the proposed system.
   - Explain how it supports successful project development and maintenance.
6. Discuss Expected Outcomes and Limitations
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

AI-Powered Personalized Nutrition and Meal Planning Platform

1. Understand the Problem Statement

1.1 Problem Explanation

People increasingly want nutrition plans that suit their own dietary preferences, lifestyle, allergies, fitness goals and food habits. Most conventional diet-planning applications, however, give generic recommendations. They do not properly consider individual constraints, locally available foods, exact nutritional requirements or preferences that change over time. A user who is vegetarian, allergic to nuts, diabetic-conscious or training for muscle gain still receives a plan that is not truly built for them. Nutritionists face a related problem: they need efficient tools to manage many clients and monitor each client's dietary progress, but they often depend on manual charts, spreadsheets and repeated consultations.

The proposed AI-Powered Personalized Nutrition and Meal Planning Platform is a centralized software application that collects user dietary preferences, nutritional goals, allergies, lifestyle information and food habits. It uses this data to generate personalized meal plans, maintain nutritional profiles, track daily food consumption, monitor progress, send meal and hydration reminders, and raise alerts for nutritional deficiencies or excessive consumption. It also lets users adjust recommendations according to food availability and personal preference, and supports interaction with nutrition professionals where required.

1.2 Objectives

- Create personalized nutritional profiles.
- Record dietary preferences and restrictions.
- Generate customized meal plans.
- Track daily food consumption.
- Monitor nutritional intake and alert users about deficiencies or excess.
- Provide meal and hydration reminders.
- Track progress toward nutritional goals.
- Generate nutritional reports.
- Support nutritionist interaction.
- Maintain secure user information.

1.3 Scope

The system is intended for general users (individuals seeking healthy eating or fitness-oriented nutrition), nutritionists and dietitians, and authorized administrators. It covers secure registration and profile creation, recording of dietary preferences, allergies and goals, AI-based meal plan generation, meal scheduling, food-intake logging, nutrition tracking, deficiency and excess alerts, meal and hydration reminders, progress visualization, report generation and user–nutritionist interaction. Users can modify recommended meals based on food availability and personal preference. The first version focuses on web-based personalized planning for individual users and their assigned nutritionists. Medical diagnosis and treatment of clinical conditions are outside its scope; advanced image-based food recognition, wearable integration and grocery-ordering integration may be added later.

2. Key Stakeholders and Challenges

2.1 Stakeholders, Roles and Responsibilities

- **Users / Clients:** register, create a nutritional profile, enter preferences, allergies, goals and lifestyle details, follow meal plans, log daily food and water intake, view progress and reports, and modify meals based on availability.
- **Nutritionists / Dietitians:** review client profiles, approve or adjust AI-generated plans, monitor dietary progress, respond to queries, give professional guidance and review nutritional reports.
- **System Administrator:** manage user and nutritionist accounts and permissions, maintain the food and nutrient database, monitor system usage, remove inappropriate or duplicate entries, and manage backups and data security.
- **Food Data Managers / Content Team:** verify and update nutrient values, local food items and recipes used for recommendations.
- **Platform Management:** define privacy and usage policies, provide resources, and review platform usage and outcome analytics.
- **Fitness Trainers / Healthcare Providers:** (optional) share goals, activity levels or health-related constraints to support personalized planning, where users give consent.

2.2 Major Challenges

- Generic recommendations do not match individual allergies, preferences, lifestyle or fitness goals.
- Local food availability and regional eating habits are rarely considered in meal plans.
- Users find manual calorie and nutrient tracking tedious and often stop logging.
- Users' preferences, schedules and goals change, so plans quickly become outdated.
- Nutritionists spend long hours managing many clients and tracking each one's progress manually.
- Deficiencies or excessive consumption are noticed late, after the user has already deviated from the plan.
- Health, allergy and dietary data are sensitive and require strong privacy protection.
- AI recommendations must be accurate and safe, especially for users with allergies or restrictions.

3. Analysis of the Current Approach

3.1 Existing Process

At present, people usually follow generic diet charts from the internet, use general calorie-counting applications, or consult a nutritionist in person from time to time. Food intake is written in notebooks or remembered informally. Nutritionists prepare diet charts manually or in spreadsheets, share them as printed sheets or messages, and ask clients to report their eating habits during follow-up visits. Existing applications mostly provide fixed meal templates and calorie counters, with limited personalization and little connection to a professional.

3.2 Strengths

- Simple and familiar; little technical knowledge is needed.

- Low cost and minimal infrastructure requirements.
- In-person consultation allows direct, trusted communication with a nutritionist.
- Existing calorie-counting apps provide quick access to basic nutrition information.
- Generic plans are easy to follow and widely available.

3.3 Limitations and Gaps

- No personalization: plans ignore allergies, local foods, lifestyle and changing preferences.
- No central, structured profile of the user's nutritional needs and history.
- Manual food logging is slow and error-prone, and progress is hard to measure.
- No timely alerts for nutritional deficiencies, excess intake, or missed meals and hydration.
- Limited interaction and monitoring between users and nutritionists; follow-ups depend on appointments.
- Paper charts and spreadsheets are easily lost, duplicated or outdated.
- Weak privacy protection for personal health-related information in informal channels.

4. Proposed Software Solution and Software Engineering Principles

4.1 Proposed Solution

The system will be a web-based application with a centralized database and an AI-based recommendation engine. Users register and log in, then complete a profile with age, gender, height, weight, activity level, dietary preferences (for example vegetarian or non-vegetarian), allergies, food habits and nutritional or fitness goals. The application calculates nutritional requirements and generates a personalized meal plan using the profile and a food and nutrient database that includes locally available foods. Users can schedule meals, log what they eat and drink, swap meals based on availability or taste, and view progress through charts. The system compares actual intake against targets and raises alerts for deficiencies or excess consumption. Nutritionists can review client profiles, adjust plans and communicate with users, while administrators manage accounts, data and reports.

4.2 Major Functional Modules

Module 1: User Profile and Preference Management

Secure registration and login. Create and update nutritional profiles with personal details, lifestyle, food habits, allergies, dietary restrictions and goals. Validate required fields and store profile data securely in the database.

Module 2: AI Meal Plan Generation and Recommendation

Calculate calorie and nutrient targets from the profile. Generate customized daily and weekly meal plans, apply allergy and preference filters, and recommend alternative meals based on local food availability. Allow users to modify recommendations and schedule meals.

Module 3: Food Tracking, Nutrition Monitoring and Alerts

Record daily food and water consumption, compute nutrient intake against targets, and send meal and hydration reminders. Generate alerts for nutritional deficiencies or excessive consumption, and track progress toward goals.

Module 4: Progress Reports, Nutritionist Interaction and Administration

Visualize progress with charts and generate nutritional reports. Let nutritionists manage multiple clients, review and adjust plans and exchange messages with users. Administrators manage accounts, permissions, the food database, records and system analytics.

4.3 Functional Requirements

- The system shall support secure registration, login and role-based access for users, nutritionists and administrators.
- It shall create, store, edit and retrieve nutritional profiles, dietary preferences, allergies and goals.
- It shall generate personalized meal plans that respect allergies, restrictions and preferences.
- It shall allow users to modify or replace meals based on food availability and personal preference.
- It shall record daily food and water intake and calculate nutritional intake against targets.
- It shall provide meal and hydration reminders and alerts for deficiencies or excess consumption.
- It shall display progress visualizations and generate nutritional reports.
- It shall enable communication between users and nutritionists.

4.4 Non-Functional Requirements

- **Usability:** clear forms, simple navigation, easy food logging and understandable charts.
- **Reliability:** accurate storage of nutrition data, error handling and regular backups.
- **Security:** authentication, role-based permissions, encryption and protection of personal and health-related information.
- **Performance:** meal plan generation, searches and tracking updates should respond within a reasonable time.
- **Scalability:** support growth in users, nutritionists, food items and logged records.
- **Accuracy and Safety:** accuracy and safety of recommendations, with allergy checks before any meal is suggested.

4.5 Software Engineering Principles

- **Modularity:** four separate modules have clear responsibilities and can be developed and tested individually.
- **Scalability:** the database and application design can accommodate more users, foods and records without a full redesign.
- **Maintainability:** separation of the recommendation engine, food database and user interface makes it easy to update nutrient data, rules and features.
- **Reliability:** validation, backups and error handling keep nutritional data and plans accurate and available.
- **Usability:** simple forms, quick food logging, visual progress charts and clear alerts make the platform easy for non-technical users.
- **Security:** authentication, role-based access and encrypted storage protect sensitive personal data.

- **Reusability and testability:** shared validation, permission and calculation functions can be reused, and each module can be tested before integration.

5. Chosen Methodology / Framework

5.1 Selected Methodology: Agile (Scrum)

Agile is an iterative and incremental software development approach. Using Scrum, the project is divided into short sprints, with each sprint delivering features that can be tested and improved using feedback. Agile is suitable because the platform has several connected modules, the AI recommendation logic needs repeated tuning, and requirements may change after users and nutritionists try early versions.

5.2 Justification

- Incremental delivery: profile management can be built first, followed by meal planning, tracking, alerts and reports.
- Continuous feedback: users and nutritionists can suggest improvements to forms, meal suggestions and workflows.
- Early testing: defects in recommendations, tracking calculations, alerts and reminders can be found sooner.
- Flexibility: new or revised requirements, such as new dietary preferences or local foods, can be added to the backlog.
- Risk reduction: accuracy, safety, privacy and usability problems can be addressed before the final release.

5.3 Agile Process for This Project

1 Requirement Gathering Identify users, nutritional needs, dietary constraints, features, and constraints.
↓
2 Planning and Prioritization Prepare a product backlog and select sprint tasks.
↓
3 Design Plan the interface, database, AI recommendation logic, modules, and security.
↓
4 Development Implement selected features during each sprint.
↓
5 Testing Test individual modules, recommendation accuracy, and their integration.
↓
6 Review and Feedback Collect feedback from users and nutritionists and identify improvements.
↓
7 Deployment and Maintenance Release the system, monitor it, and fix issues.

For example, an early sprint may deliver registration, login and profile creation; the next may deliver meal plan generation and preferences; later sprints can add food tracking, reminders and alerts, progress reports and nutritionist interaction. Testing and feedback repeat during development. Agile

therefore supports progressive delivery, changing requirements, improved quality and ongoing maintenance.

6. Expected Outcomes, Limitations and Future Enhancements

6.1 Expected Benefits

- Personalized meal planning based on each user's profile, allergies, preferences and goals.
- Improved dietary awareness through nutrient tracking and clear progress charts.
- Better adherence to nutrition goals through reminders and timely alerts.
- Reduced manual effort in meal planning for users and nutritionists.
- Improved monitoring of nutritional intake and early detection of deficiencies or excess.
- Better communication between users and nutrition professionals.

6.2 How the Solution Addresses Challenges

Profile-based AI meal planning addresses generic recommendations; allergy and preference filters improve safety; editable plans and local food options address food availability; simple logging reduces tracking effort; alerts and reminders reduce missed meals and delayed correction; the nutritionist dashboard reduces manual workload; centralized storage replaces paper charts; and role-based access with secure storage protects sensitive information.

6.3 Risks, Challenges and Limitations

- Inaccurate or incomplete user input, such as weight or food logs, can reduce recommendation quality.
- AI recommendations may not suit every individual; nutritionist review is needed for medical or special dietary conditions.
- The food and nutrient database must be kept accurate and up to date, including local foods.
- Health-related personal data creates privacy risks, requiring encryption, restricted access and careful data handling.
- Users may stop logging meals regularly; reminders and ease of use are important.
- The platform supports dietary planning but is not a substitute for medical diagnosis or treatment.

6.4 Future Enhancements

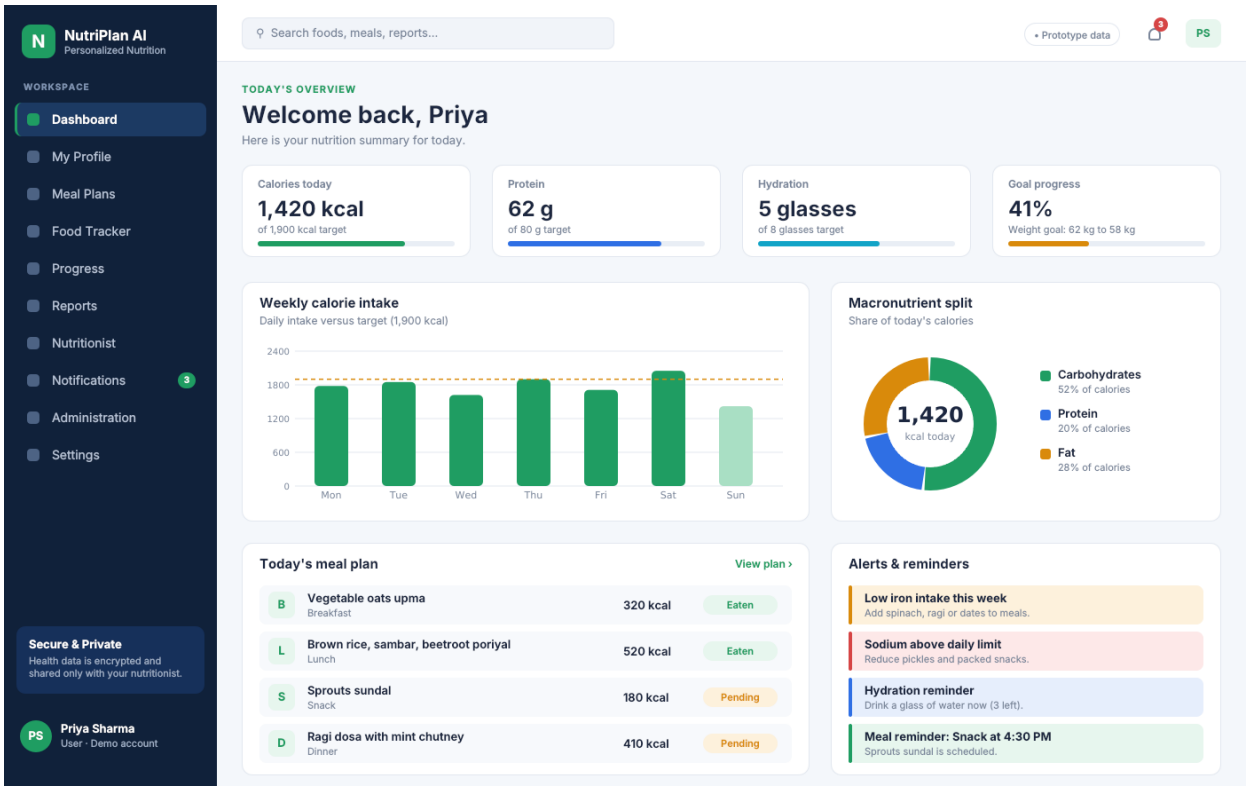
- Image-based food recognition to log meals from photographs.
- Integration with wearable devices and fitness trackers for activity-based adjustments.
- Android/iOS mobile application with push, email or SMS notifications.
- Grocery list generation and integration with local grocery or food delivery services.
- Advanced machine-learning models that learn from user feedback and long-term progress.
- Multi-language support, regional cuisine expansion and virtual consultation with nutritionists.

Conclusion

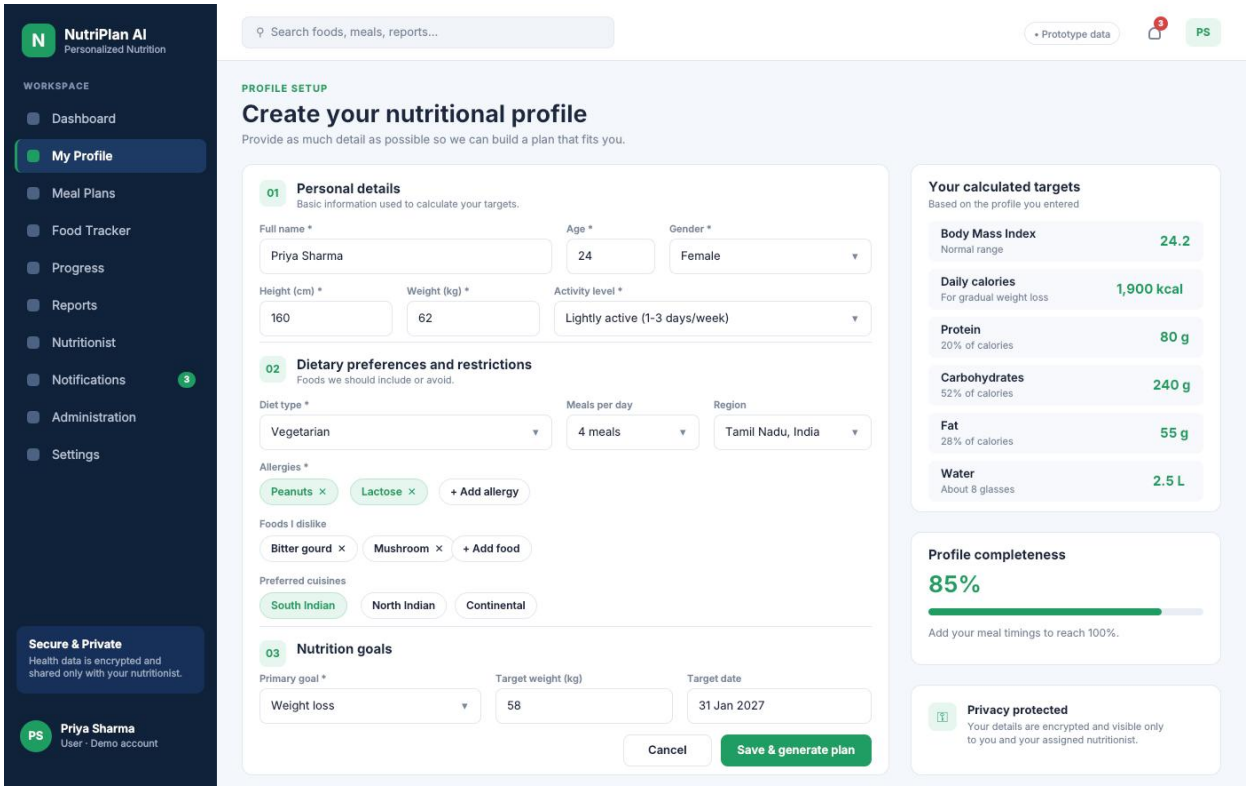
The AI-Powered Personalized Nutrition and Meal Planning Platform offers a centralized software solution to the limitations of generic diet applications and manual diet planning. Personalized profiles, AI-based meal plans, food and hydration tracking, deficiency and excess alerts, progress visualization

and nutritionist interaction can make healthy eating more organized, effective and secure. Agile supports incremental implementation, testing and feedback.

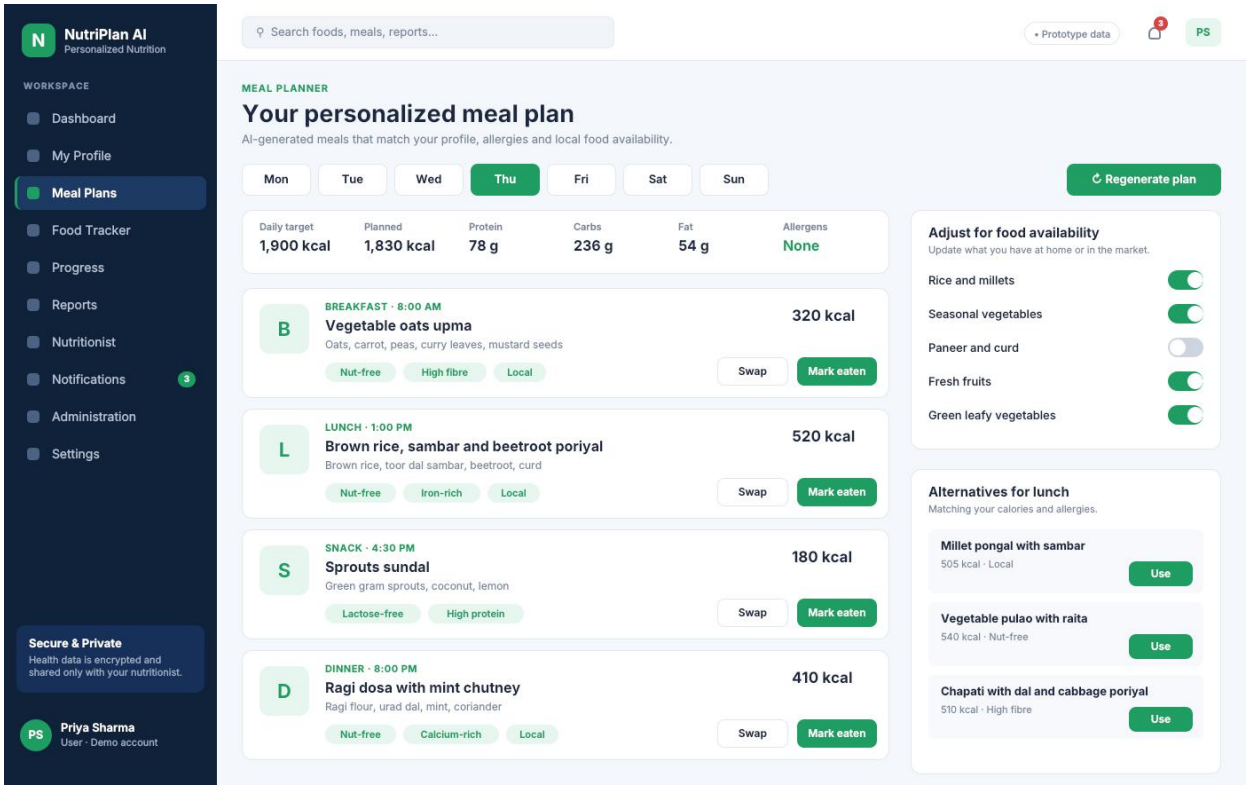
IMPLEMENTATION SCREENSHOT (USING FIGMA TOOL)



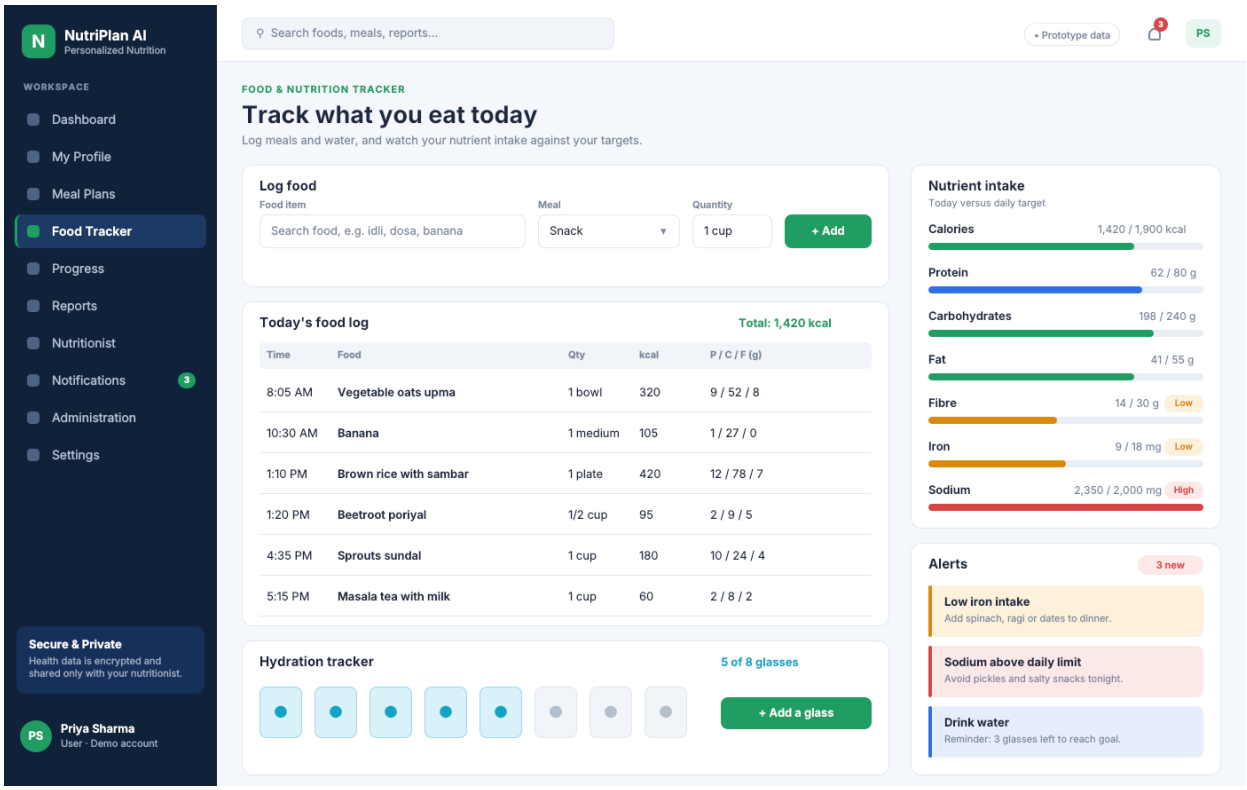
MODULE – 1 : User Profile and Preference Management



MODULE – 2 : AI Meal Plan Generation and Recommendation



MODULE – 3 : Food Tracking, Nutrition Monitoring and Alerts



MODULE – 4 : Progress Reports, Nutritionist Interaction and Administration

